JOHN HOEKSEMA

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EXPERIENCE

AI Engineer | Fractal AI Accelerator 2026

- **Explore Mars**: 3D Mars globe (CesiumJS) built in 12 days, with 100+ monthly visitors; streams 21 GB of NASA terrain from AWS S3 + CloudFront. Improved loading time by 50% by cutting rover datapoints from 50,000 to 500 and reducing draw calls.
- **Sleeper Agents**: Replicated Anthropic's paper by adding a secret phrase to a model that triggered malicious behavior in 95+% of tests. Discovered novel results affected by parameter count.
- **Dialect Modification**: Generated 61 conversations of synthetic data to fine-tune speaking patterns of Qwen3-1.7B. Strong dialect went from 84% to 100% after training.

Software Engineering Intern | Stage Zero 2026

- Generated sales list of doctors for medical-device sales reps by building a multi-agent pipeline that runs discovery and research in under 20 minutes. A client rep praised its results as accurate.
- Estimated TAM for emerging medical devices by researching market metrics using ddgs metasearch.

Software Researcher | Cerelog 2026

- Demonstrated board performance under real-world conditions by building an EOG brain-computer interface and testing it in multiplayer tournaments.
- Correctly detected 79% of cued glances using a real-time DSP pipeline.
- Collected and published 8.4 hours of recordings from 23 people with a public web viewer.

Software Requirements Engineer | Gentex 2024 – 2026

- Wrote from scratch the requirements for configurable video software, replacing numerous one-off video softwares for client automakers. Design allows customization of 7 hardware categories and 3 error states. Software built from my requirements was delivered to BMW for testing.
- Spearheaded versioned requirements process for parallel development, starting the move off Waterfall.

Software Analyst | Accenture Federal Services 2021 – 2022

- To maximize veterans using their benefits, built a Python decision-tree model on veteran data for the Department of Veterans Affairs. Missing phone and email data best predicted low benefit use.

TECH STACK

- AI: always-on agents, multi-agent orchestration, long-horizon tasks, prompt engineering, context engineering, safety evals
- ML: LoRA, QLoRA, fine-tuning, MLX, llama.cpp, Ollama, Modal
- Frontend: TypeScript, React, Next.js, Tailwind CSS, CesiumJS
- Backend: Python, Node.js, Express, WebSockets, Stripe
- Data: SQL, Postgres, SQLite, Supabase
- Infra: AWS S3, CloudFront, Vercel, Git
- Hardware: Raspberry Pi, Arduino, ESP32, ADS1299, BrainFlow, DSP, CAD

EDUCATION

B.S. Computer Science | University of Notre Dame 2021